

Arc type after DFS, arc $u \rightarrow v$

- **Tree:** in the search forest. **Forward:** not tree, u ancestor of v . **Back:** u descendant of v . **Cross:** neither is an ancestor.
- Tree/forward: $s[u] < s[v] < d[v] < d[u]$ (tree iff $\text{pred}[v] = u$, else forward).
- Back: $s[v] < s[u] < d[u] < d[v]$.
- Cross: $s[v] < d[v] < s[u] < d[u]$.
- Never possible: $s[a] < s[b] < d[a] < d[b]$ (intervals nest or are disjoint).
- When arc first explored: v white $\Rightarrow$ tree, v grey $\Rightarrow$ back, v black $\Rightarrow$ forward or cross.
- Acyclic **iff** DFS finds no back arcs. Root is last to turn black; children finish before parent.
- Graph (undirected): no edges between different trees.

Topological order

- Exists **iff** DAG. Not unique in general.
- **DFS:** no back arcs $\Rightarrow$ list nodes by **decreasing done time**. (Every arc has $d[u] > d[v]$.)
- **Zero-in-degree:** repeatedly remove a source. No source left but nodes remain $\Rightarrow$ cycle.
- Unique iff Hamiltonian path.[†] Max size of DAG of order n : $n(n-1)/2$.[†] With in-degree ≤ 2 : node i has $\min(i-1, 2)$ in-arcs, $n=5$: $0+1+2+2+2=7$.[†]
- Topological order $\Rightarrow$ directed girth $= \infty$, but girth can still be finite (e.g. $0 \rightarrow 1, 1 \rightarrow 2, 0 \rightarrow 2$).

Girth

- **Girth** = shortest cycle of the *underlying graph* (∞ if none). A cycle in a graph has length ≥ 3 (uvu is not a cycle).
- **Directed girth** = shortest directed cycle. A 2-cycle $u \rightleftarrows v$ counts here (directed girth 2) but is a single edge in the underlying graph.
- A directed cycle of length ≥ 3 is also a cycle of the underlying graph, so $\text{girth} \leq \text{directed girth}$ *unless directed girth* = 2 (then girth can be bigger or ∞).
- **Undirected, BFS from v :** on meeting a grey neighbour, finish that level, then stop. If v is the LCA of x, y : cycle length $d(x) + d(y) + 1$. Take the min.
- **Directed, BFS from v :** first back arc (x, v) gives $1 + d(x)$. Stop at once.
- Girth or directed girth of whole graph: run from every node, $\Theta(n+m)$ with lists.
- DFS back arcs give a cycle, but may miss the shortest.

Walk, path, cycle (L20)

- **Walk:** arc between each consecutive pair. **Path:** no repeated node. **Cycle:** walk with $v_0 = v_l$, no other repeats. Length = number of arcs; walk/path can have length 0.
- Check each step: right *direction*, and cycle needs length ≥ 3 in a graph.
- uvu : walk yes, path no, cycle no (graph).
- Directed cycle present $\Rightarrow$ longest walk unbounded.[†] DAG: longest walk = longest path $\leq n-1$.
- **Distance $d(u, v)$:** arcs on a shortest path, ∞ if none.
- **Induced subdigraph on V' :** all arcs with both ends in V' . Size = arcs inside V' .
- **Reverse:** flip every arc. **Underlying graph:** drop directions, merge $u \rightleftarrows v$ into one edge.
- Source: in-degree 0. Sink: out-degree 0.
- Order $n = |V|$, size $m = |E|$. Digraph: $0 \leq m \leq n(n-1)$, no loops. **Sparse:** $m = O(n)$. **Dense:** $m = \Omega(n^2)$.
- $\sum \text{deg} = 2m$.[†] Min vertices for m edges, no isolated: smallest k , $\binom{k}{2} \geq m$ (45 edges $\Rightarrow k = 10$).[†]

Connectivity and components (L26)

- Connected graph: path between every pair. **Components** = trees of a BFS/DFS forest = number of roots chosen, $O(n+m)$.
- Digraph, BFS/DFS gives one tree $\Rightarrow$ underlying graph connected. It does *not* prove strongly connected.
- **Strongly connected:** every pair reachable both ways.
- Components of graph, order n size m :[†] **min** $\max(1, n-m)$. **Max:** smallest k with $\binom{k}{2} \geq m$, then $n-k+1$.
- Order 12 size 5: $k=4$, max 9. Order 8 size 9: $k=5$, max 4, min 1.[†]

BFS (L24)

- Queue. Grey on discovery. Black when dequeued, after all neighbours examined. **Root first black; parent turns black before its children; black order = discovery order.**
- Arrays pred (tree parent) and $d[v] = d(r, v)$ for root r of v 's tree.
- Tiebreak: lowest index. Queue empty $\Rightarrow$ restart at lowest-index white node (new tree).
- **Follow arc direction:** only out-neighbours are explored.
- **Digraph:** tree arc $d[w] = d[v] + 1$. Back arc: w black, $d[w] \leq d[v] - 1$. **No forward arcs.** Cross arc: $d[w] \leq d[v]$. So no non-tree arc goes deeper.
- **Graph:** tree edge $|\Delta d| = 1$. Cross edge $\Delta d = 0$ or $|\Delta d| = 1$.
- Any cycle forces a non-tree edge $\Rightarrow$ cross edge exists (graph). 4-cycle from A: A0, B1, D1, C2, leftover C–D joins levels 2, 1. 5-cycle: leftover joins levels 2, 2. Even cycle $\Rightarrow$ different levels.

PFS and worked examples

- **PFS:** choose grey node with smallest key. Key = time turned grey $\Rightarrow$ BFS. Key = $-\text{seen} \Rightarrow$ DFS. Array version $\Theta(n^2)$.
- **BFS order = DFS order, one tree each:** strongest conclusion is underlying graph *connected*. Star $r \rightarrow a, b, c$ and directed path both qualify, so not a path, not diameter ≤ 2 , no node needs out-degree $n-1$.
- **Third black node:** if BFS from 2 reaches only $\{2, 1\}$ then restart at lowest white node 0, giving 2, 1, 0.

Adjacency matrix vs lists (L21)

	Matrix	Lists
arc (i, j) exists?	$\Theta(1)$	$\Theta(d)$
outdegree of i	$\Theta(n)$	$\Theta(1)$
indegree of i	$\Theta(n)$	$\Theta(n + m)$
add arc	$\Theta(1)$	$\Theta(1)$
delete arc	$\Theta(1)$	$\Theta(d)$
add node	$\Theta(n)$	$\Theta(1)$
delete node	$\Theta(n^2)$	$\Theta(n + m)$
space	$\Theta(n^2)$	$\Theta(n + m \log n)$
BFS/DFS	$\Theta(n^2)$	$\Theta(n + m)$

d = outdegree of i . Lists are doubly linked.

- Matrix row i = out-arcs of i (outdegree = number of 1s in row). Column j = in-arcs of j .
- **Reverse digraph:** matrix transposes. Row of reverse = column of original. Reverse lists (and all indegrees) from lists in $\Theta(n + m)$.
- Space: lists better when sparse, comparable when dense. Matrix competitive when dense.
- Access middle of list: array $\Theta(1)$, linked list $\Theta(n)$. Sequential search $\Theta(n)$.

Traversal running times

- One BFS/DFS over whole graph (restarts included): $\Theta(n + m)$ lists, $\Theta(n^2)$ matrix.
- BFS/DFS *from every node* (no skipping): $n \times \Theta(n + m)$. Sparse $\Theta(n^2)$, dense with lists $\Theta(n^3)$.
- Arc pass + BFSvisit from each node + linear extras, sparse lists: $\Theta(n^2)$.
- Reverse digraph then DFS from every node: $\Theta(n + m) + n \Theta(n + m)$.
- Components: $O(n + m)$.

Selection and BST (L14–L16)

- **Quickselect:** partition, recurse into *one* side. Average $\Theta(n)$, worst $\Theta(n^2)$ (bad pivots, same as quicksort). Quicksort: avg $n \log n$, worst n^2 .
- **BST:** every key in left subtree $\leq$ node $\leq$ every key in right subtree (equal keys allowed). Check against *all* ancestors. In-order is sorted.
- Operations $\Theta(h)$, worst $h = n - 1$. Random inserts: average $\Theta(\log n)$.
- Delete: two children $\Rightarrow$ replace by min of right subtree (or max of left), then delete that node.
- **AVL:** balance = $h(\text{right}) - h(\text{left}) \in \{-1, 0, 1\}$. $h \leq 1.44 \lg n$. Insert 4,3,2,1 gives chain, one right rotation fixes it.[†]
- **Red-black:** root black, leaves black, red node has black children, same black count on every path. $h \leq 2 \lg(n + 1)$, max depth $\leq 2 \times$ min depth.
- **Binary search:** $\Theta(\log n)$ with $\Theta(1)$ access (array), not on a linked list. $T(n) = 1 + T(\lfloor n/2 \rfloor)$.

Table (worst/avg)	Insert	Find	Delete
Sorted array	n	$\log n$	n
BST	$n / \log n$	$n / \log n$	$n / \log n$
Balanced BST	$\log n$	$\log n$	$\log n$
Hash, chaining	n / λ	n / λ	n / λ

Hashing (L17–L19)

- Load factor $\lambda = n/m$. Collision: two keys, same slot. Uniform hashing model (balls in bins).
- **Chaining:** insert $\Theta(1)$, delete $\Theta(1)$ (doubly linked). Unsuccessful search λ , successful $\approx 1 + \lambda/2$. Worst $\Theta(n)$. Fine if λ bounded.
- **Open addressing:** all keys in table. Linear probing moves **left** one slot, wrapping. Double hashing: step $p(k)$ left. Linear probing clusters.
- Insert cost = unsuccessful search. Uniform/double hashing: $\Theta(\frac{1}{1-\lambda})$. Linear probing: $\Theta(1 + \frac{1}{(1-\lambda)^2})$. Resize as $\lambda \rightarrow 1$.
- Linear probing, Knuth:[†] hit $\frac{1}{2}(1 + \frac{1}{1-\lambda})$, miss $\frac{1}{2}(1 + \frac{1}{(1-\lambda)^2})$. At $\lambda = 0.9$: 5.5 and 50.5.
- Deletion in open addressing needs a tombstone, else later searches stop early.[†]
- Balls in bins: empty fraction $\approx e^{-\lambda}$ (expected empty bins $m(1 - \frac{\lambda}{m})^n$). First collision after $\approx \sqrt{\pi m/2}$ balls. All bins non-empty after $\approx m \ln m$. Fullest bin $\approx \log n / \log \log n$.
- **Universal family:** $\Pr[h(k) = h(k')] \leq 1/m$. Example $h_{ab}(x) = ((ax + b) \bmod p) \bmod m$, $p \geq m$ prime.
- Fullest bin for $h(x) = x \bmod 10$: group keys by last digit.